Class 11

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Figure 18

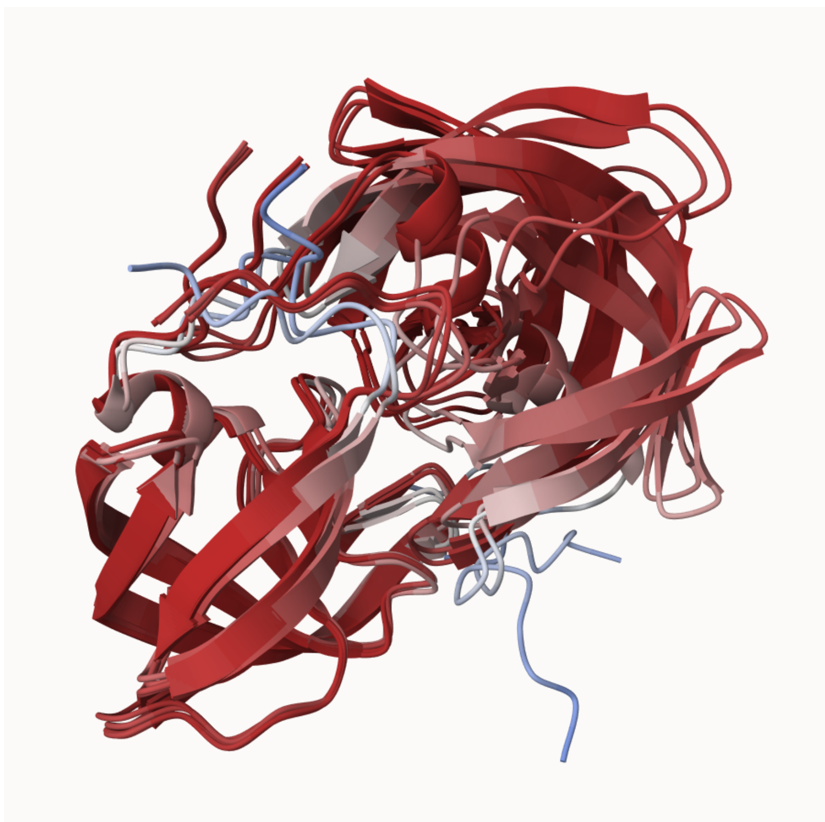


Figure: Superposed monomer models

8. Custom analysis of resulting models

```r
# Set the folder with PDB model files
results_dir <- "~/Downloads/hivprdimer_23119/"
```

```r
# Get all PDB files from the results folder
pdb_files <- list.files(path=results_dir,
                        pattern="*.pdb",
                        full.names = TRUE)
# Print PDB file names
basename(pdb_files)
```

```
[1] "hivprdimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "hivprdimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "hivprdimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "hivprdimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "hivprdimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```r
library(bio3d)

# Read all data from Models
#  and superpose/fit coords
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
/Users/leascy/Downloads/hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_001_alphafold2_mul
/Users/leascy/Downloads/hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_002_alphafold2_mul
/Users/leascy/Downloads/hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_003_alphafold2_mul
/Users/leascy/Downloads/hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_004_alphafold2_mul
/Users/leascy/Downloads/hivprdimer_23119//hivprdimer_23119_unrelaxed_rank_005_alphafold2_mul
.....

Extracting sequences

pdb/seq: 1   name: /Users/leascy/Downloads/hivprdimer_23119//hivprdimer_23119_unrelaxed_rank
pdb/seq: 2   name: /Users/leascy/Downloads/hivprdimer_23119//hivprdimer_23119_unrelaxed_rank
pdb/seq: 3   name: /Users/leascy/Downloads/hivprdimer_23119//hivprdimer_23119_unrelaxed_rank
pdb/seq: 4   name: /Users/leascy/Downloads/hivprdimer_23119//hivprdimer_23119_unrelaxed_rank
pdb/seq: 5   name: /Users/leascy/Downloads/hivprdimer_23119//hivprdimer_23119_unrelaxed_rank
```

```r
#Reading and aligning PDB structures
pdbs
```

```
                              1         .         .         .         .         50
[Truncated_Name:1]hivprdimer   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]hivprdimer   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]hivprdimer   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]hivprdimer   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]hivprdimer   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
                               **************************************************
                               1         .         .         .         .         50

                              51         .         .         .         .         100
[Truncated_Name:1]hivprdimer   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:2]hivprdimer   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:3]hivprdimer   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:4]hivprdimer   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:5]hivprdimer   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
                               **************************************************
                              51         .         .         .         .         100

                             101         .         .         .         .         150
[Truncated_Name:1]hivprdimer   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:2]hivprdimer   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:3]hivprdimer   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:4]hivprdimer   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:5]hivprdimer   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
                               **************************************************
                             101         .         .         .         .         150

                             151         .         .         .         .       198
[Truncated_Name:1]hivprdimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]hivprdimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]hivprdimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]hivprdimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]hivprdimer   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
                               ************************************************
                             151         .         .         .         .       198

Call:
  pdbaln(files = pdb_files, fit = TRUE, exefile = "msa")

Class:
  pdbs, fasta

Alignment dimensions:
```

```
  5 sequence rows; 198 position columns (198 non-gap, 0 gap)

+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

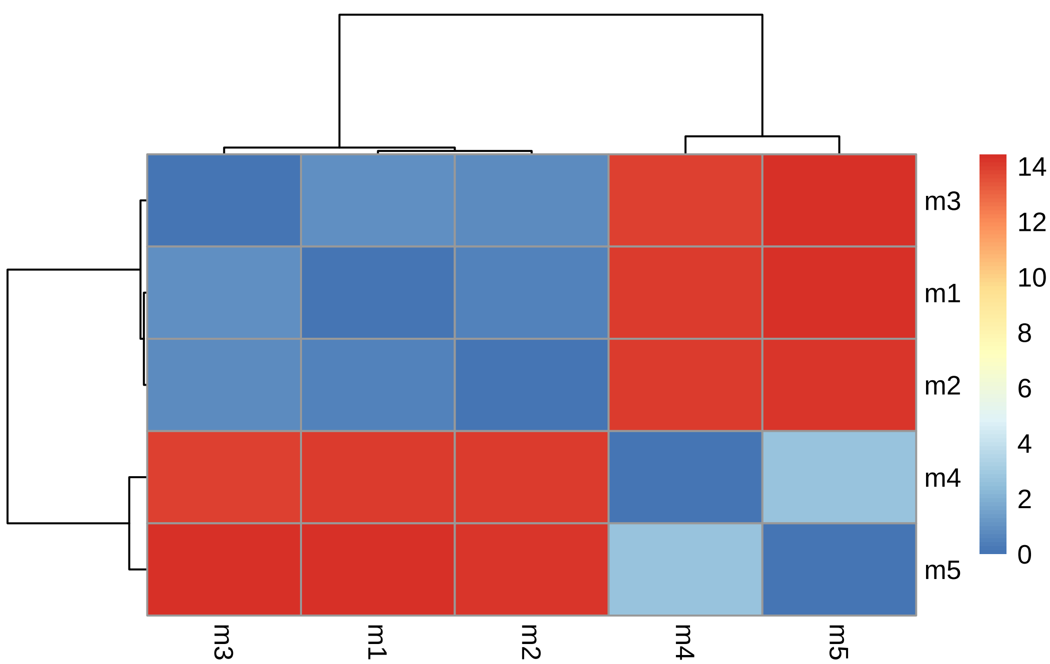
```r
#Calculating the RMSD betweena all pairs models with rmsd()
rd <- rmsd(pdbs, fit=TRUE)
```

```
Warning in rmsd(pdbs, fit = TRUE): No indices provided, using the 198 non NA positions
```

```r
#Showing the range of RMSD values between models
range(rd)
```

```
[1]  0.000 14.428
```

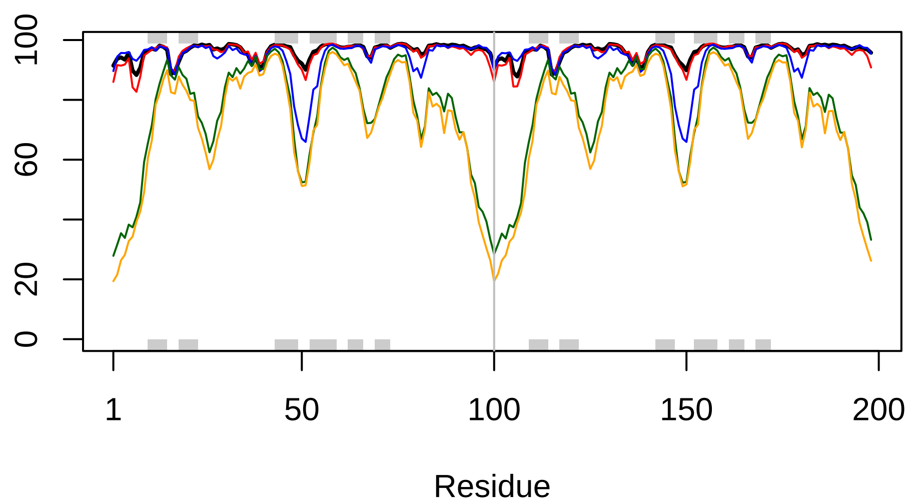
```r
#Generating heatmap of RMSD matrix values
#Loading the pheatmap package
library(pheatmap)
#Assigning column names to each model
colnames(rd) <- paste0("m",1:5)
#Assigning row names to each model
rownames(rd) <- paste0("m",1:5)
#Plotting the RMSD heatmap
pheatmap(rd)
```



```
# Read a reference PDB structure
pdb <- read.pdb("1hsg")
```

```
## Note: Accessing on-line PDB file
```

```
# Ploting B-factor/flexibility values for model 1
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
# Adding model 2 B-factor values to the plot
points(pdbs$b[2,], typ="l", col="red")
#Adding model 3 B-factor values to the plot
points(pdbs$b[3,], typ="l", col="blue")
#Adding model 4 B-factor values to the plot
points(pdbs$b[4,], typ="l", col="darkgreen")
#Adding model 5 B-factor values to the plot
points(pdbs$b[5,], typ="l", col="orange")
#Adding a vertical reference line at residue position 100
abline(v=100, col="gray")
```




```
#Improving supersition/fitting of models by finding the most "rigid core" common
#Across all the models
core <- core.find(pdbs)
```

```
core size 197 of 198  vol = 8545.071
core size 196 of 198  vol = 7895.678
core size 195 of 198  vol = 3578.372
core size 194 of 198  vol = 1851.165
core size 193 of 198  vol = 1697.196
core size 192 of 198  vol = 1612.763
core size 191 of 198  vol = 1530.195
core size 190 of 198  vol = 1447.395
core size 189 of 198  vol = 1377.104
core size 188 of 198  vol = 1303.813
core size 187 of 198  vol = 1239.028
core size 186 of 198  vol = 1188.127
core size 185 of 198  vol = 1118.462
core size 184 of 198  vol = 1071.664
core size 183 of 198  vol = 1034.006
core size 182 of 198  vol = 980.826
core size 181 of 198  vol = 942.239
core size 180 of 198  vol = 911.387
core size 179 of 198  vol = 879.749
core size 178 of 198  vol = 834.459
core size 177 of 198  vol = 785.261
core size 176 of 198  vol = 762.115
core size 175 of 198  vol = 722.023
core size 174 of 198  vol = 700.389
core size 173 of 198  vol = 677.251
core size 172 of 198  vol = 657.804
core size 171 of 198  vol = 632.902
core size 170 of 198  vol = 614.193
core size 169 of 198  vol = 591.511
core size 168 of 198  vol = 573.974
core size 167 of 198  vol = 552.398
core size 166 of 198  vol = 529.484
core size 165 of 198  vol = 500.54
core size 164 of 198  vol = 482.513
core size 163 of 198  vol = 458.422
core size 162 of 198  vol = 444.451
core size 161 of 198  vol = 433.577
core size 160 of 198  vol = 419.082
core size 159 of 198  vol = 404.931
core size 158 of 198  vol = 393.799
core size 157 of 198  vol = 382.999
core size 156 of 198  vol = 366.651
core size 155 of 198  vol = 352.023
```

```
core size 154 of 198  vol = 335.659
core size 153 of 198  vol = 319.395
core size 152 of 198  vol = 307.932
core size 151 of 198  vol = 296.815
core size 150 of 198  vol = 284.286
core size 149 of 198  vol = 273.457
core size 148 of 198  vol = 261.975
core size 147 of 198  vol = 249.597
core size 146 of 198  vol = 237.951
core size 145 of 198  vol = 226.098
core size 144 of 198  vol = 213.263
core size 143 of 198  vol = 200.212
core size 142 of 198  vol = 187.502
core size 141 of 198  vol = 177.523
core size 140 of 198  vol = 167.371
core size 139 of 198  vol = 160.874
core size 138 of 198  vol = 154.453
core size 137 of 198  vol = 148.438
core size 136 of 198  vol = 142.129
core size 135 of 198  vol = 136.528
core size 134 of 198  vol = 130.768
core size 133 of 198  vol = 123.867
core size 132 of 198  vol = 117.608
core size 131 of 198  vol = 112.709
core size 130 of 198  vol = 106.36
core size 129 of 198  vol = 100.59
core size 128 of 198  vol = 95.717
core size 127 of 198  vol = 91.067
core size 126 of 198  vol = 86.861
core size 125 of 198  vol = 82.309
core size 124 of 198  vol = 78.554
core size 123 of 198  vol = 74.632
core size 122 of 198  vol = 70.489
core size 121 of 198  vol = 66.802
core size 120 of 198  vol = 62.901
core size 119 of 198  vol = 59.152
core size 118 of 198  vol = 55.75
core size 117 of 198  vol = 51.831
core size 116 of 198  vol = 48.3
core size 115 of 198  vol = 44.926
core size 114 of 198  vol = 42.417
core size 113 of 198  vol = 39.425
core size 112 of 198  vol = 37.381
```

```
 core size 111 of 198  vol = 33.06
 core size 110 of 198  vol = 28.153
 core size 109 of 198  vol = 25.33
 core size 108 of 198  vol = 22.509
 core size 107 of 198  vol = 20.695
 core size 106 of 198  vol = 18.754
 core size 105 of 198  vol = 17.757
 core size 104 of 198  vol = 16.712
 core size 103 of 198  vol = 15.44
 core size 102 of 198  vol = 14.745
 core size 101 of 198  vol = 14.758
 core size 100 of 198  vol = 13.11
 core size 99 of 198  vol = 11.018
 core size 98 of 198  vol = 8.967
 core size 97 of 198  vol = 7.643
 core size 96 of 198  vol = 6.326
 core size 95 of 198  vol = 5.37
 core size 94 of 198  vol = 4.312
 core size 93 of 198  vol = 3.391
 core size 92 of 198  vol = 2.697
 core size 91 of 198  vol = 1.911
 core size 90 of 198  vol = 1.577
 core size 89 of 198  vol = 1.144
 core size 88 of 198  vol = 0.826
 core size 87 of 198  vol = 0.594
 core size 86 of 198  vol = 0.494
 FINISHED: Min vol ( 0.5 ) reached
```

```r
#Write out the fitted structures to corefit_structures directory
core.inds <- print(core, vol=0.5)
```

```
# 87 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     8  50     43
2    52  95     44
```

```r
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

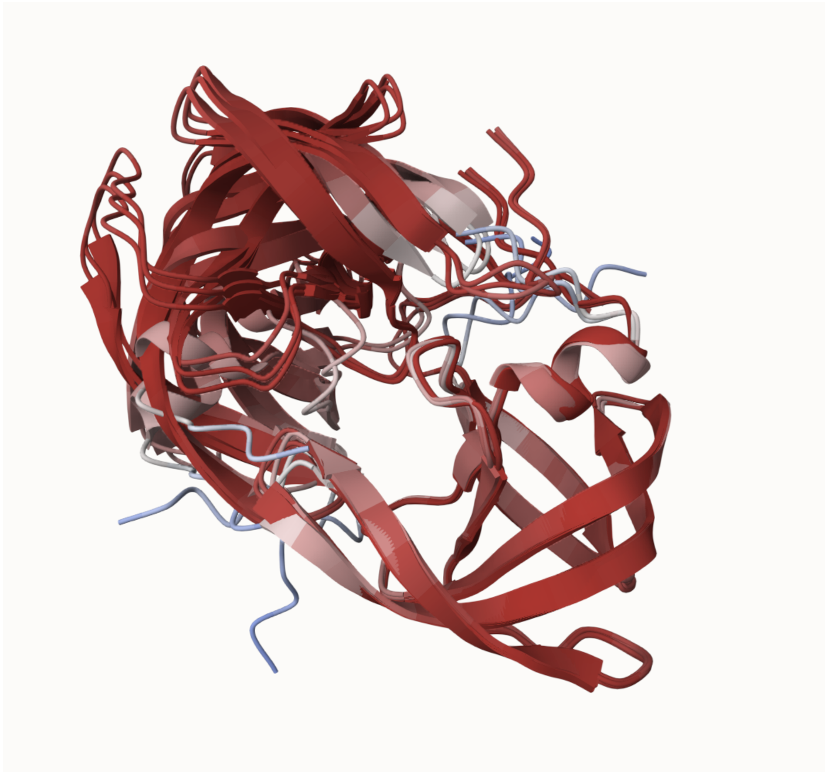
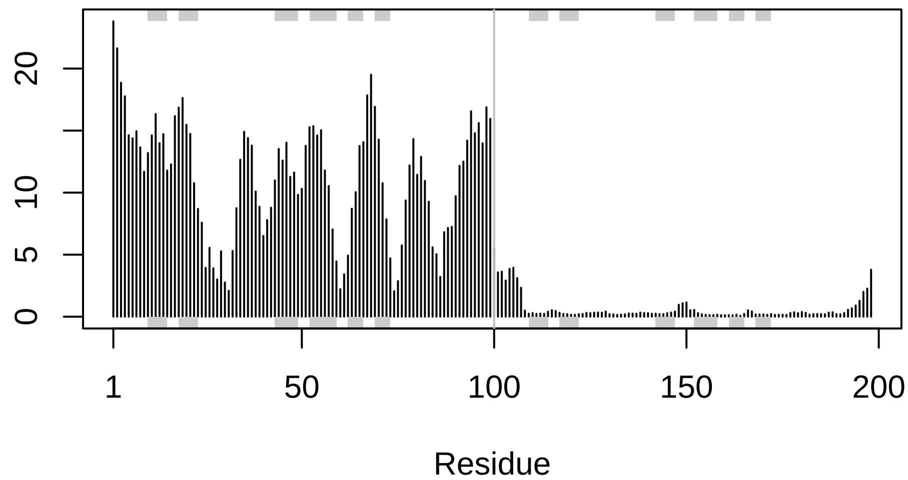


Figure 19

```
rf <- rmsf(xyz)

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```



Predicted Alignment Error for domains

```
library(jsonlite)

# Listing of all PAE JSON files
pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)
```

```
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

```
# Per-residue pLDDT scores
#  same as B-factor of PDB..
head(pae1$plddt)
```

```
[1] 91.44 93.75 94.19 93.44 95.62 89.81
```

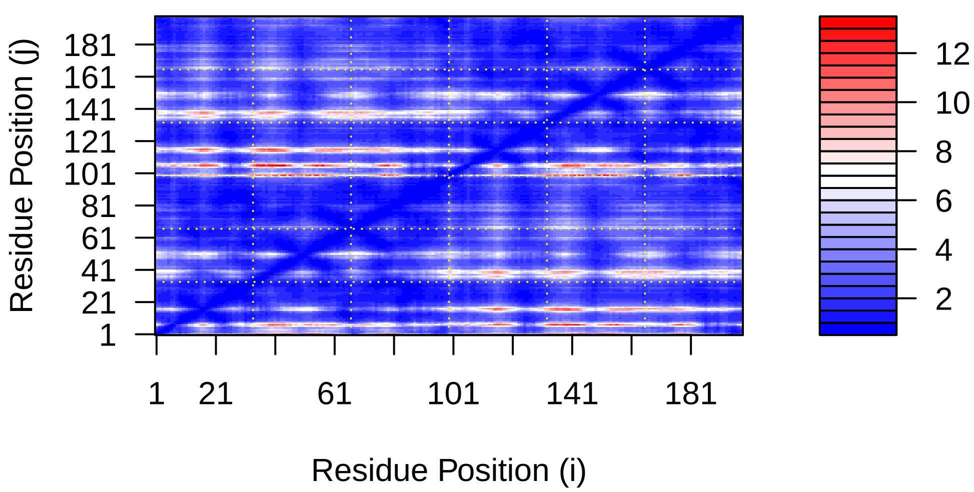
```
pae1$max_pae
```

```
[1] 13.13281
```

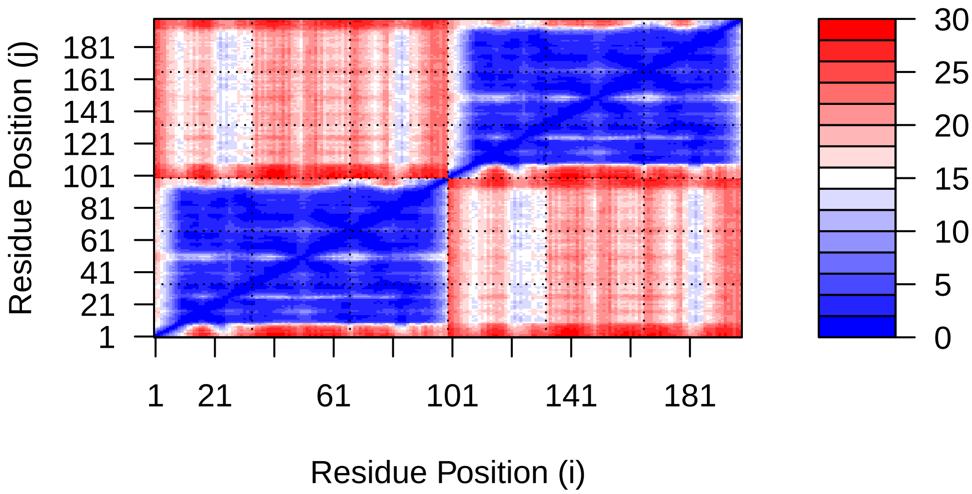
```
pae5$max_pae
```

```
[1] 29.71875
```

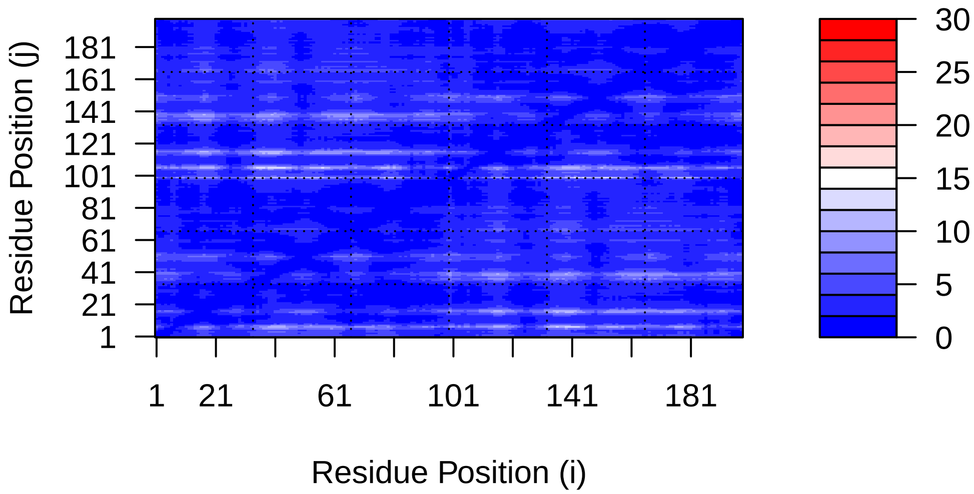
```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```




```
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```




```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Residue conservation from alignment file

```
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                        full.names = TRUE)
aln_file
```

```
[1] "/Users/leascy/Downloads/hivprdimer_23119//hivprdimer_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

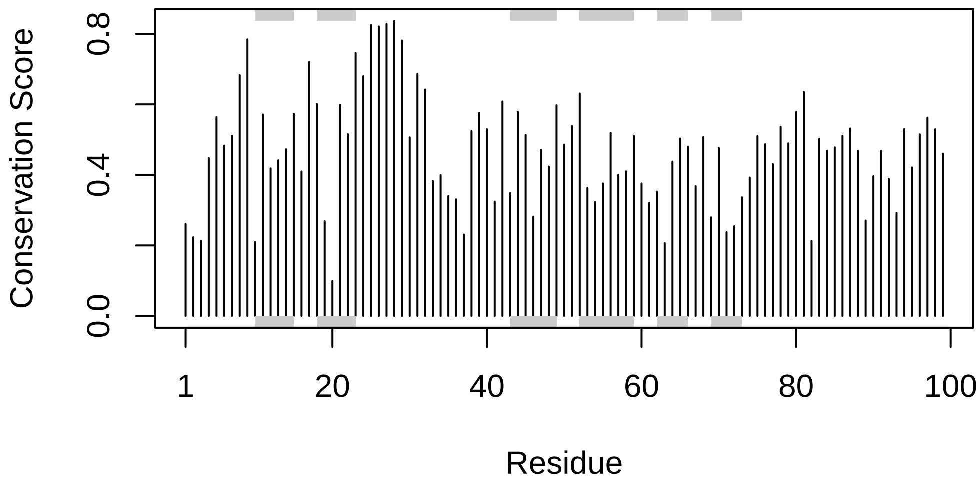
```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

```
[1] 5397  132
```

```
sim <- conserv(aln)
```

```r
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```




```r
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```r
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```

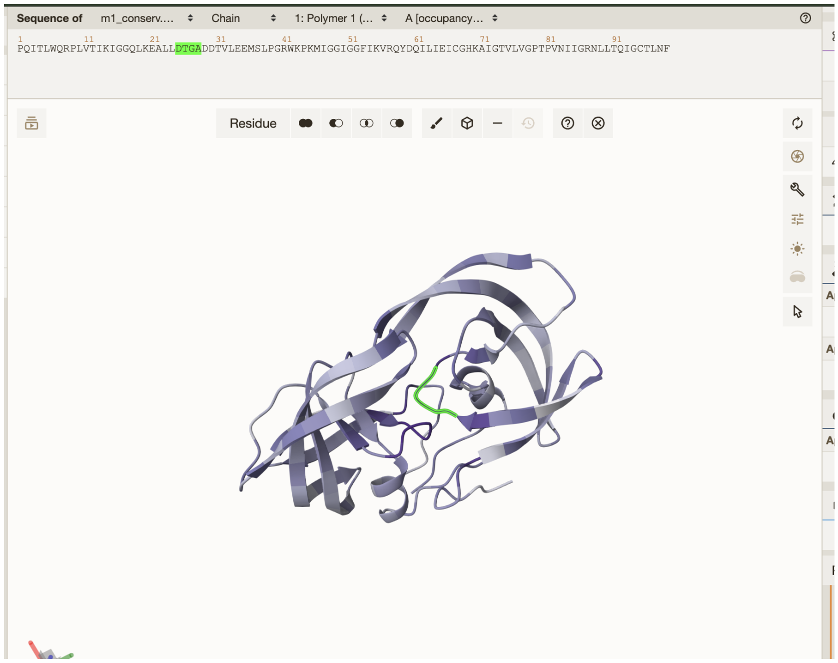

Sequence of
m1_conserv....
Chain
1: Polymer 1 (...
A [occupancy...
1 11 21 31 41 51 61 71 81 91
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
Residue

Figure 20